Tutorial-15

Modules name:Pycharm
Test Case ID:p_15

Name: Selenium Environment setup

1. **open the pycharm**
2. **write the code**
3. **open the terminal**
4. **click on run**

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
NoSuchElementException,
TimeoutException,
ElementNotInteractableException,
StaleElementReferenceException
)
URL = "https://www.saucedemo.com/"
V ALID_USERNAME = "standard_user"
V ALID_PASSWORD = "secret_sauce"
def start_browser():
driver = webdriver.Chrome()
driver.maximize_window()
return driverdef login(driver, username, password):
driver.get(URL)
wait = WebDriverWait(driver, 10)
username_box = wait.until(
EC.visibility_of_element_located(
(By.ID, "user-name")
)
)
password_box = wait.until(
EC.visibility_of_element_located(
(By.ID, "password")
)
)
login_button = wait.until(
EC.element_to_be_clickable(
(By.ID, "login-button")
)
)
username_box.clear()
```

```
username_box.send_keys(username)
password_box.clear()
password_box.send_keys(password)
login_button.click()
# TC-01 Valid Login
def test_valid_login():
driver = start_browser()try:
login(
driver,
V ALID_USERNAME,
V ALID_PASSWORD
)
WebDriverWait(driver, 10).until(
EC.url_contains("inventory")
)
assert "inventory" in driver.current_url
products_title = WebDriverWait(
driver, 10
).until(
EC.visibility_of_element_located(
(By.CLASS_NAME, "title")
)
)
assert products_title.text == "Products"
print("TC-01 Valid Login: PASS")
print("Authentication Assertion: PASS")
except AssertionError:
print("TC-01 Valid Login: FAIL")
except NoSuchElementException:
print("NoSuchElementException handled")
except TimeoutException:
print("TimeoutException handled")
except ElementNotInteractableException:
print(
"ElementNotInteractableException handled"
)except StaleElementReferenceException:
print(
"StaleElementReferenceException handled"
)
finally:
driver.quit()
# TC-02 Invalid Password
def test_invalid_password():
driver = start_browser()
try:
login(
driver,
V ALID_USERNAME,
"wrong_password"
)
error_message = WebDriverWait(
driver, 10
).until(
EC.visibility_of_element_located(
```

```
(
By.CSS_SELECTOR,
"h3[data-test='error']"
)
)
)
assert (
"Username and password do not match"
in error_message.text
)
print("TC-02 Invalid Password: PASS")except AssertionError:
print("TC-02 Invalid Password: FAIL")
except TimeoutException:
print("Error message was not displayed")
finally:
driver.quit()
# TC-03 Invalid Username
def test_invalid_username():
driver = start_browser()
try:
login(
driver,
"wrong_user",
V ALID_PASSWORD
)
error_message = WebDriverWait(
driver, 10
).until(
EC.visibility_of_element_located(
(
By.CSS_SELECTOR,
"h3[data-test='error']"
)
)
)
assert (
"Username and password do not match"
in error_message.text
)print("TC-03 Invalid Username: PASS")
except AssertionError:
print("TC-03 Invalid Username: FAIL")
except TimeoutException:
print("Error message was not displayed")
finally:
driver.quit()
# TC-04 Empty Credentials
def test_empty_credentials():
driver = start_browser()
try:
driver.get(URL)
login_button = WebDriverWait(
driver, 10
).until(
EC.element_to_be_clickable(
```

```
(By.ID, "login-button")
)
)
login_button.click()
error_message = WebDriverWait(
driver, 10
).until(
EC.visibility_of_element_located(
(
By.CSS_SELECTOR,
"h3[data-test='error']"
)
))
assert (
"Username is required"
in error_message.text
)
print("TC-04 Empty Credentials: PASS")
except AssertionError:
print("TC-04 Empty Credentials: FAIL")
except TimeoutException:
print("Validation message was not displayed")
finally:
driver.quit()
# TC-05 Logout
def test_logout():
driver = start_browser()
try:
login(
driver,
V ALID_USERNAME,
V ALID_PASSWORD
)
WebDriverWait(driver, 10).until(
EC.url_contains("inventory")
)
menu_button = WebDriverWait(
driver, 10
).until(
EC.element_to_be_clickable((By.ID, "react-burger-menu-btn")
)
)
menu_button.click()
logout_button = WebDriverWait(
driver, 10
).until(
EC.element_to_be_clickable(
(By.ID, "logout_sidebar_link")
)
)
logout_button.click()
WebDriverWait(driver, 10).until(
EC.visibility_of_element_located(
(By.ID, "login-button")
```

)
)
assert "saucedemo.com" in driver.current_url
print("TC-05 Logout: PASS")
except AssertionError:
print("TC-05 Logout: FAIL")
except TimeoutException:
print("Logout operation timed out")
finally:
driver.quit()
Execute all test cases
test_valid_login()
test_invalid_password()test_invalid_username()
test_empty_credentials()
test_logout()

Output:

